

Cascadia Classifier Model Retraining Pipeline

Use Case 1 – Model Maintainer

The Cascadia Classifier UI has a Feedback tab, intended to help identify misclassified images and images not present in the dataset. The general process for integrating the feedback into a retrained model is as follows: gather the landmarks specified in the feedback; add the landmarks to the landmarks_washington_full.csv dataset; put the landmark images into a folder, named after the landmark; add the folder to data/images; and retrain the model.

1. Gather the landmarks specified in the feedback. Each feedback form is emailed to the Cascadia Classifier email address.

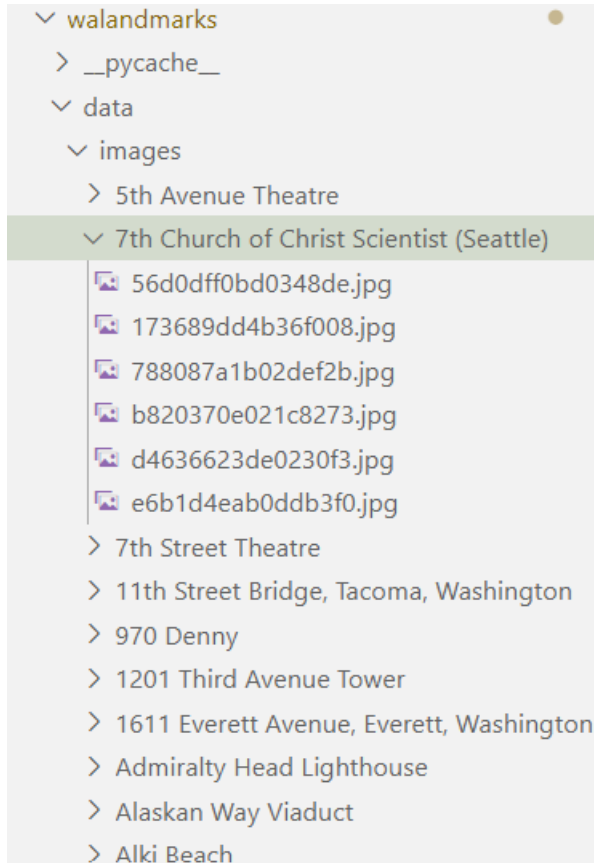
☐	☆	me	New message from - Hello! There is a new submission from the WA Landmarks Classification feedback form. Submitted by: Ann...	2:00 PM
☐	☆	me	New message from - Hello! There is a new submission from the WA Landmarks Classification feedback form. Submitted by: Ann...	1:58 PM
☐	☆	me	New message from - Hello! There is a new submission from the WA Landmarks Classification feedback form. Submitted by: Ann...	1:55 PM
☐	☆	me	New message from - Hello! There is a new submission from the WA Landmarks Classification feedback form. Submitted by: Ann...	1:22 PM
☐	☆	me	New message from - Hello! There is a new submission from the WA Landmarks Classification feedback form. Submitted by: Ann...	1:11 PM
☐	☆	me	New message from - Hello! There is a new submission from the WA Landmarks Classification feedback form. Submitted by: Anni...	12:57 PM
☐	☆	me	New message from - Hello! There is a new submission from the WA Landmarks Classification feedback form. Submitted by: Anni...	12:23 PM
☐	☆	me	New message from - Hello! There is a new submission from the WA Landmarks Classification feedback form. Submitted by: Dem...	12:15 PM
☐	☆	me	New message from - Hello! There is a new submission from the WA Landmarks Classification feedback form. Submitted by: Izzy ...	11:07 AM
☐	☆	me	New message from - Hello! There is a new submission from the WA Landmarks Classification feedback form. Submitted by: Anni...	10:23 AM

2. Add the landmarks to the landmarks_washington_full.csv dataset. If the landmark is new to the dataset, assign it a new landmark_id.

1	landmark_id,name,supercategory,location,latitude,longitude,category
2	161,Lime Kiln Point State Park,Washington state park,"San Juan Island, San Juan County,
3	239,Illahee State Park,Washington state park,"Kitsap County, Washington, Pacific Northwe
4	262,Comet Falls,waterfall,"Mount Rainier National Park, Lewis County, Washington, Pacifi
5	291,"St. Andrews Episcopal Church (Chelan, Washington)",church building,"Washington, Pac
6	517,Titlow Beach,landform,"Tacoma, Pierce County, Washington, Pacific Northwest, Washing
7	763,St. Demetrios Greek Orthodox Church (Seattle),church building,"Washington, Pacific N
8	769,Beckler Peak,mountain,"King County, Washington, United States",47.73649722222226,-1
9	798,"Granite Mountain (King County, Washington)",mountain,"King County, Washington, Unit
10	1139,Lake Whatcom,lake,"Washington, Whatcom County, United States",48.73305555555556,-12
11	1255,"Bryant, Seattle, Washington",neighborhood,"Seattle, King County, Washington, Pacif
12	1528,Dome Peak,mountain,"Washington, Pacific Northwest, Washington, Pacific States Regio
13	1554,Potholes Reservoir,reservoir,"Washington, Pacific Northwest, Washington, Pacific St
14	1633,"Lincoln Theatre (Mount Vernon, Washington)",movie theater,"Mount Vernon, Skagit Co
15	1711,"Broadmoor, Seattle, Washington",neighborhood,"Washington, Pacific Northwest, Washi
16	2393,Steptoe Battlefield State Park,Washington state park,"Rosalia, Whitman County, Wash

3. Put the landmark images into a folder, named after the landmark.

4. Add the folder to data/images. The final structure after steps (3) and (4) is shown below.



5. Retrain the model. To do this, run the file `walandmarks/helpers/landmark_classification_model_training.py`. The main guard of this file calls the function `create_train_analyze_model`. This function takes five parameters: `save_model_flag`, a Boolean flag indicating whether to save the model; `save_model_path_name`, the filepath at which to save the model (used if `save_model_flag` is True); `loss_curve_path_name`, the path at which to save the loss curve; `accuracy_curve_path_name`, the path at which to save the accuracy curve; and `auc_curve_path_name`, the path at which to save the AUC curve. The function returns None but creates four files:
 - a. The trained model, saved at `save_model_path_name`
 - b. A plot of the train and validation loss by epoch
 - c. A plot of the train and validation accuracy by epoch
 - d. A plot of the train and validation AUC by epoch.